

Statistics (1)

Syllabus

Yu-You Liou

Fall 2026

Lecturer: Yu-You Liou	Contact: d10627008@ntu.edu.tw
Course No.: EIB-21E-01-A1	Website: https://yyliou.github.io/stat
Meeting Time: Tuesday, Periods 8–9 (15:10–17:00)	Location: L309

Course Objectives

1. To train students in basic statistical knowledge and econometric concepts.
2. To enhance students' familiarity with data analysis and provide hands-on experience in statistical analysis.

Textbooks

1. Bluman, A. G. (2012). *Elementary Statistics: A Step-by-Step Approach* (8th ed.). McGraw-Hill.

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **Attendance** (12%): $R = \min\{12, t\}$, where t represents the number of sessions attended.
2. **In-Class Exercises** (48%): 12 exercises $\times$ 4% each, submitted at the end of each session.
3. **Mid-Term Examination** (20%) and **Final Oral Examination** (20%).

Regulations

1. Attendance

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. Examination

Students who do not show up for their scheduled examination will receive a grade of zero for that component.

3. In-class exercise deadline

The deadline for all in-class exercises is before midnight of the following week. Late submissions will not be accepted.

4. Academic Integrity

All submitted work must be the student's own. Copying from other students is strictly prohibited. AI detection tools are used to review submissions; any work found to be copied will receive a grade of zero.

5. AI Policy

Students may use AI tools (e.g., ChatGPT, GitHub Copilot) as learning aids, but must be able to explain and defend any code or analysis they submit. Uncritical submission of AI-generated work without understanding constitutes academic dishonesty.

Required Software

All software is free and open-source: **R** at <https://cran.r-project.org>, **RStudio Desktop** at <https://posit.co/download/rstudio-desktop>, and R packages **tidyverse**, **ggplot2**, **patchwork** (installation instructions on the course R page).

Course Schedule

Week	Topic
1	Explain the syllabus and class regulations
2	Chapter 1: The nature of probability and statistics (ex1)
3	Chapter 2: Frequency distributions and graphs (ex2)
4	Chapter 3: Data description (ex3)
5	Chapter 3: Data description (ex4)
6	Chapter 4: Probability and counting rules (ex5)
7	Chapter 4: Probability and counting rules (ex6)
8	Midterm review
9	Midterm exam
10	Chapter 5: Discrete probability distributions (ex7)
11	Chapter 5: Discrete probability distributions (ex8)
12	Chapter 6: The normal distribution (ex9)
13	Chapter 6: The normal distribution (ex10)
14	Chapter 7: Confidence intervals and sample size (ex11)
15	Chapter 7: Confidence intervals and sample size (ex12)
16	Final exam
17	Flexible learning week
18	Flexible learning week

Note: Syllabus is subject to change. Any updates will be announced on the course website.